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EYS-162

B. Tech. (New Scheme) Examination, 2024

(Fourth Semester)

NUMERICAL ANALYSIS

EL-401

Time : 3 Hours]

[Maximum Marks : 60

Note : All questions are compulsory and carry equal marks. Attempt any two parts from each question.

1. (a) Find the positive root of $f(x) = x^3 - x - 1$ correct to four decimal places by Bisection method.
- (b) Using Newton-Raphson method find the root between 0 and 1 of $f(x) = x^3 - 6x + 4$ correct to 5 decimal places.

- (c) Using Lagrange's Interpolation formula, find $y(10)$ from the following table :

x	5	6	9	11
y	12	13	14	16

2. (a) Evaluate the integral :

$$I = \int_4^{5.2} \log_e x \, dx$$

using Trapezoidal and Simpson's $\frac{1}{3}$ as well as

Simpson's $\frac{3}{8}$ rule.

- (b) Find, by Gaussian elimination method the inverse of :

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

- (c) Solve the following system by Gauss-Jacobi method :

$$\begin{aligned} 10x - 5y - 2z &= 3; & 4x - 10y + 3z &= -3; \\ x + 6y + 10z &= -3. \end{aligned}$$

3. (a) Solve :

$$\frac{dy}{dx} = x + y, \text{ given } y(1) = 0 \text{ and get } y(1.1), y(1.2)$$

by Taylor series method.

- (b) Apply the fourth order Runge-Kutta method to find $y(0.2)$ given that $y' = x + y$, $y(0) = 1$.

- (c) Solve :

$$\frac{dy}{dx} = y - \frac{2x}{y}; y(0) = 1 \text{ for } y(0.1), \text{ taking } h = 0.1$$

and using Modified Euler's method.

4. (a) Explain Binomial Distribution. Hence find the probability of getting 4 heads in 6 tosses of a fair coin.

- (b) If the variance of the Poisson distribution is 2, find the probabilities for $r = 1, 2, 3, 4$ from the recurrence relation of the Poisson distribution. Also find $P(r \geq 4)$.

(c) Prove that :

$$L[\cosh at] = \frac{s}{s^2 - a^2}$$

5. (a) If $L[F(t)] = f(s)$, then prove that
 $L[e^{at}F(t)] = f(s - a)$
- (b) Find the Laplace transform of the periodic function (sawtooth wave) $f(t) = \frac{K}{T}t; 0 < t < T$ and $F(t + T) = F(t)$
- (c) State and prove convolution theorem.